

Customer Shopping Behavior Analysis

A data-driven review of retail purchase patterns, customer segments, and subscription trends

1. Overview

This report examines shopping behavior across 3,900 retail transactions to understand how customers spend, which segments generate the most value, and how factors such as discounts, shipping preferences, and subscription status shape purchasing decisions. The findings are intended to inform strategic decisions around marketing, retention, and product positioning.

2. Dataset at a Glance

The underlying dataset consists of **3,900 transaction records** described by **18 attributes**, grouped into three broad categories:

- Demographics** — age, gender, location, and subscription status
- Transaction details** — item purchased, category, purchase amount, season, size, and color
- Behavioral signals** — discount usage, promo code usage, purchase history, purchase frequency, review rating, and shipping type

One data quality issue was identified during profiling: the Review Rating field was missing 37 values, which was resolved in the cleaning stage described below.

3. Data Preparation and Exploration (Python)

Before analysis, the raw dataset was cleaned and enriched using pandas. The process followed these steps:

- Loading the data** — the raw CSV was read into a pandas DataFrame for processing.
- Profiling the structure** — `df.info()` and `.describe()` were used to understand data types, distributions, and summary statistics across all fields.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	1
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	1
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Summary statistics generated with `.describe()` — part 1

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

Summary statistics generated with .describe() — part 2

- **Handling missing values** — the 37 missing Review Rating entries were filled using the median rating within each product category, preserving category-level rating patterns.
- **Standardizing column names** — all fields were converted to snake_case for consistency and easier downstream querying.
- **Engineering new features** —
 - an age_group field was derived by binning customer ages into cohorts;
 - a purchase_frequency_days field was derived from the recorded purchase cadence.
- **Checking for redundancy** — discount_applied and promo_code_used were found to carry overlapping information, so promo_code_used was dropped from further analysis.
- **Loading into a database** — the cleaned DataFrame was pushed into MySQL, setting up the dataset for SQL-based analysis.

4. Business Analysis (SQL)

With the cleaned data in MySQL, ten targeted queries were written to answer specific business questions. Each query and its output are shown below.

1. **Revenue split by gender.** How does total revenue compare between male and female shoppers?

	gender	revenue
▶	Male	157890
	Female	75191

2. **Discount users who still spend big.** Which customers redeemed a discount yet still spent above the overall average purchase amount? (839 matching records)

	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79

3. **Best-rated products.** Which five products earned the highest average review scores?

	item_purchased	Average Product Rating
►	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

4. **Standard vs. Express shipping.** Do customers who choose Express shipping spend more, on average, than those who choose Standard?

	shipping_type	Round
►	Express	60.48
	Standard	58.46

5. **Subscribers vs. non-subscribers.** How do customer counts, average spend, and total revenue compare across subscription status?

	subscription_status	total_customers	avg_spend	total_revenue
►	Yes	1053	59.49	62645
	No	2847	59.87	170436

6. **Products most reliant on discounts.** Which five products have the highest share of discounted purchases?

	item_purchased	discount_rate
►	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

7. **Customer segmentation.** How many customers fall into the New, Returning, and Loyal segments based on purchase history?

	customer_segment	Number of Customers
►	Loyal	3116
	Returning	701
	New	83

8. **Category leaders.** Within each product category, which three items are ordered most often?

	item_rank	category	item_purchased	total_orders
►	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

9. **Repeat purchasing and subscription uptake.** Are customers with more than five purchases more likely to hold an active subscription?

	subscription_status	repeat_buyers
►	Yes	958
	No	2518

10. **Revenue by age cohort.** How much total revenue does each age group contribute?

	age_group	total_revenue
▶	young adults	62143
	middle-aged	59197
	adult	55978
	senior	55763

5. Interactive Dashboard (Power BI)

To make these findings easy to explore, the results were also assembled into an interactive Power BI dashboard. It brings together customer counts, average purchase amount, review ratings, subscription mix, category-level revenue and sales, and age-group breakdowns in a single filterable view.



6. Recommendations

Based on the analysis above, the following actions are recommended:

- **Grow the subscriber base** — subscribers currently make up a minority of customers; offering exclusive perks could encourage more sign-ups.
- **Invest in loyalty** — reward repeat purchasers to convert more Returning customers into the high-value Loyal segment.

- **Reassess discount strategy** — several products depend heavily on discounting to sell, which may be compressing margins and warrants review.
- **Lean into what's already working** — feature top-rated and best-selling items more prominently in campaigns and merchandising.
- **Sharpen marketing targeting** — prioritize the age groups and shipping segments that generate the most revenue, such as Express-shipping customers.